

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Differentiates 2^t or 2^{-t} to obtain $\pm A \ln 2 \times 2^{\pm t}$	AO1.1a	M1	
	Obtains $\frac{dy}{dt} = (\pm A \ln 2)2^t$ and $\frac{dx}{dt} = (\pm B \ln 2)2^{-t}$	AO1.1b	A1	$\frac{dy}{dt} = (3 \ln 2)2^t$ $\frac{dx}{dt} = (-4 \ln 2)2^{-t}$
	Uses chain rule with correct $\frac{dy}{dt}$ and $\frac{dx}{dt}$ and completes rigorous argument to obtain fully correct printed answer	AO2.1	R1	$\frac{dy}{dx} = \frac{(3 \ln 2)2^t}{(-4 \ln 2)2^{-t}}$ $= -\frac{3}{4} \times 2^{2t}$
(b)	Rearranges to write 2^{-t} in terms of x or 2^t in terms of y Or Writes given expression in terms of t	AO3.1a	M1	$2^t = \frac{y+5}{3}$ $2^{-t} = \frac{x-3}{4}$ $1 = \left(\frac{y+5}{3}\right)\left(\frac{x-3}{4}\right)$
	Eliminates t Or Compares coefficients PI by $a = 5$ or $b = -3$	AO1.1a	M1	$12 = xy + 5x - 3y - 15$ $xy + 5x - 3y = 27$
	Completes rigorous argument to obtain correct values of a , b and c and write the Cartesian equation in the required form ISW	AO2.1	R1	ALT $xy + ax + by = (4 \times 2^{-t} + 3)(3 \times 2^t - 5) + a(4 \times 2^{-t} + 3) + b(3 \times 2^t - 5)$ $= 12 - 15 + (4a - 20)2^{-t} + (3b + 9)2^t + 3a - 5b$ $a = 5, b = -3$ $xy + 5x - 3y = -3 + 15 + 15$ $= 27$
				Total 6 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses a correct method for finding $\frac{dy}{dx}$ evidence for this includes sight of $\frac{dy}{dx}$ or $\frac{dx}{dt}$ and chain rule OR an attempt at implicit or explicit differentiation of a correct Cartesian equation or 'their' equation from part (b)	AO1.1a	M1	$\frac{dx}{dt} = 3t^2$ $\frac{dy}{dt} = 2t$ $\frac{dy}{dx} = \frac{2t}{3t^2}$ When $t = -2$ $\frac{dy}{dx} = -\frac{1}{3}$ ALT $y = (x-2)^{\frac{2}{3}} - 1$ $\frac{dy}{dx} = \frac{2(x-2)^{-\frac{1}{3}}}{3}$
	Obtains correct $\frac{dy}{dx}$	AO1.1b	A1	When $t = -2$, $x = -6$ $\frac{dy}{dx} = \frac{2(-6-2)^{-\frac{1}{3}}}{3} = -\frac{1}{3}$
	Substitutes $t = -2$ (or $x = -6$) into 'their' equation for $\frac{dy}{dx}$	AO1.1a	M1	
	Obtains correct simplified gradient of the curve FT 'their' equation for $\frac{dy}{dx}$	AO1.1b	A1F	
(b)	Eliminates t or makes t the subject in one expression (evidence for this includes one equation with t as the subject or two equations with equal powers of t .)	AO1.1a	M1	$t^3 = (x-2)$, $t^2 = (y+1)$ $t^6 = (x-2)^2$, $t^6 = (y+1)^3$ $(x-2)^2 = (y+1)^3$
	Finds a correct Cartesian equation in any form	AO1.1b	A1	ALT $t^3 = (x-2)$ $t = (x-2)^{\frac{1}{3}}$ $y = (x-2)^{\frac{2}{3}} - 1$
				Total 6 marks

Q3.

(a) (i) $2x - 2y \frac{dy}{dx} = 0$
Correct differentiation

M1

$$\frac{dy}{dx} = \frac{x}{y}$$

at (p, q) $\frac{dy}{dx} = \frac{p}{q}$

(p, q) substituted into correct derivative or $x = p$ $y = q$ stated

AG

A1

2

Alternative:

$$y = \sqrt{x^2 - 8} \quad \frac{dy}{dx} = \frac{1}{2} \times 2x(x^2 - 8)^{-\frac{1}{2}} = \frac{x}{y}$$

(M1)

$$= \frac{p}{q}$$

(A1)

(2)

Alternative:

$$\frac{dy}{dt} = 1 + \frac{2}{t^2} \quad \frac{dx}{dt} = 1 - \frac{2}{t^2}$$

Attempt parametric derivatives and use chain rule.

(M1)

$$\frac{dy}{dx} = \frac{1 + \frac{2}{t^2}}{1 - \frac{2}{t^2}} = \frac{t + \frac{2}{t}}{t - \frac{2}{t}} = \frac{x}{y}$$

at (p, q) $\frac{dy}{dx} = \frac{p}{q}$

(p, q) substituted into correct derivative.

(A1)

(2)

(ii) tangent at (p, q) $y - q = \frac{p}{q}(x - p)$
 ACF

B1

tangent at $(p, -q)$ $y - (-q) = \frac{-p}{q}(x - p)$
 ACF

B1

add $2y = 0$
 Solve tangent equations for y .

M1

conclusion $y = 0 \Rightarrow$ intersect on Ox
 Conclusion required

A1

4

Alternative:

tangent at (p, q) $y - q = \frac{p(x - p)}{q}$
 ACF

(B1)

tangent at $(p, -q)$ $y - (-q) = \frac{-p(x - p)}{q}$
 ACF

(B1)

When $y = 0$ $\frac{-q^2}{p} = x - p$ and $\frac{q^2}{-p} = x - p$
 Substitute $y = 0$ into both candidate's tangents
 and solve for x .

(M1)

$x = p - \frac{q^2}{p}$ is on both lines, so intersect on x axis
 Conclusion

(A1)

$$x + y = 2t \quad x - y = \frac{4}{t}$$

(4)

(b) $x^2 = t^2 + 4 + \frac{4}{t^2} \quad y^2 = t^2 - 4 + \frac{4}{t^2}$

Attempt to square x and y and subtract.

M1

$$x^2 - y^2 = 8$$

All correct AG Allow 8 = 8

A1

2

Alternative:

$$(x - y)(x + y) = 2t \times \frac{4}{t}$$

Attempt to eliminate t

(M1)

$$x^2 - y^2 = 8$$

(A1)

(2)

[8]

Q4.

(a) $\left(\frac{dx}{dt} = \right) -16e^{-2t} \quad \left(\frac{dy}{dt} = \right) 4e^{2t}$

Both derivatives correct

B1

$$\frac{dy}{dx} = \frac{\text{candidate's } \frac{dy}{dt}}{\text{candidate's } \frac{dx}{dt}}$$

chain rule used correctly

M1

$$\frac{dy}{dx} = \frac{4e^{2t}}{-16e^{-2t}} \quad \left(= -\frac{1}{4}e^{4t} \right)$$

*Simplification not required $4e^{2t}$ and $-16e^{-2t}$ must be seen.
ISW.*

		A1	3
(b) (i)	$t = \ln 2$ gradient at $P = -4$ <i>B0 if ISW result is used.</i>		
		B1ft	1
(ii)	coordinates of P $x = -2$ $y = 12$	B1	
		B1	2
(iii)	gradient of normal = $\frac{1}{4}$ <i>ft gradient at P</i> equation of normal $\frac{y-12}{x-(-2)} = \frac{1}{4}$ <i>Set up equation of normal</i>	B1ft	
		M1	
	at $y = 0$ $x = -50$ $(-50, 0)$ CSO		
		A1	3
(c)	$xy + 4y - 4x = (8e^{-2t} - 4)(2e^{2t} + 4) + 4(2e^{2t} + 4) - 4(8e^{-2t} - 4)$ <i>Write $xy + 4y - 4x$ in terms of t.</i>	M1	
	$= 16 + 32e^{-2t} - 8e^{2t} - 16 + 8e^{2t} + 16 - 32e^{-2t} + 16$ <i>Multiply out and simplify using $e^{-2t}e^{2t} = 1$ PI</i>		
		m1	
	$(xy + 4y - 4x) = 32$ <i>Correct working to $k = 32$</i> <i>$k = 32$ NMS; SC1</i>		
		A1	

Alternative:

$$e^{-2t} = \frac{x+4}{8} \text{ or } e^{2t} = \frac{y-4}{2}$$

Write e^{-2t} in terms of x or e^{2t} in terms of y .
Condone sign errors

(M1)

$$e^{-2t}e^{2t} = \left(\frac{x+4}{8}\right)\left(\frac{y-4}{2}\right) = \frac{xy+4y-4x-16}{16} = 1$$

Multiply out and use $e^{-2t} e^{2t} = 1$

(m1)

$$xy + 4y - x = 32$$

All correct with $k = 32$

(A1)

(3)

[12]

Q5.

(a) $xy^2 + 3y = (8t^2 - t)\left(\frac{3}{t}\right)^2 + 3\left(\frac{3}{t}\right)$
Substitute and expand

M1

$$= 72 - \frac{9}{t} + \frac{9}{t} = 72$$

$$k = 72$$

A1

2

Alternative

$$x = 8\left(\frac{3}{y}\right)^2 - \frac{3}{y}$$

Eliminate t

(M1)

$$xy^2 + 3y = 72$$

$$k = 72$$

(A1)

(2)

(b) (i) $\frac{dx}{dt} = 16t - 1 \quad \frac{dy}{dt} = -\frac{3}{t^2}$

B1B1

$$t = \frac{1}{4} \quad \frac{dy}{dx} = \frac{-\frac{3}{(\frac{1}{4})^2}}{16 \times \frac{1}{4} - 1}$$

Use chain rule $\left(\frac{dy}{dx} = \frac{-3}{16t^3 - t^2} \right)$ and calculate gradient
using $t = \frac{1}{4}$

M1

$$= -16$$

A1

$$t = \frac{1}{4} \quad x = \frac{8}{16} - \frac{1}{4} \quad y = \frac{3}{\frac{1}{4}}$$

Calculate x and y using $t = \frac{1}{4}$

M1

$$x = \frac{1}{4} \quad y = 12$$

Both correct

A1

tangent $y = -16x + 16$

ACF CSO

$$y - 12 = -16 \left(x - \frac{1}{4} \right) \text{ ISW}$$

A1

7

Alternative

$$2xy \frac{dy}{dx} + y^2$$

Product rule attempted; two terms added, one with $\frac{dy}{dx}$

(M1A1)

$$+ 3 \frac{dy}{dx} = 0$$

(B1)

$$t = \frac{1}{4} \quad x = \frac{8}{16} - \frac{1}{4} \quad y = \frac{3}{4}$$

Calculate x and y using $t = \frac{1}{4}$

(M1)

$$x = \frac{1}{4} \quad y = 12$$

Both correct.

(A1)

$$\left(\frac{dy}{dx} = \frac{-y^2}{2xy + 3} \right) \frac{dy}{dx} = -16$$

Calculate gradient from candidate's expression.

m1

$$\text{tangent } y = -16x + 16$$

ACF CSO

$$y - 12 = -16 \left(x - \frac{1}{4} \right) \quad \text{ISW}$$

(A1)

(7)

Alternative

$$x = \frac{72 - 3y}{y^2}$$

Correct expression for x from candidate's implicit equation.

Quotient rule attempted; y^4 and two terms subtracted.

(M1)

$$\frac{dx}{dy} = \frac{y^2(-3) - (72 - 3y) \times 2y}{y^4}$$

(A1)

$$\left(\frac{dx}{dy} = \frac{3y-144}{y^3} \right)$$

Numerator; first term; second term

(A1)

$$t = \frac{1}{4} \quad y = \frac{3}{\frac{1}{4}} = 12$$

(B1)

$$\frac{dx}{dy} = -\frac{1}{16} \quad \frac{dy}{dx} = -16$$

Use $t = \frac{1}{4}$ to calculate y

m1

$$x = \frac{1}{4} \quad x = \frac{8}{16} - \frac{1}{4} = \frac{1}{4}$$

Evaluate and invert.

(B1)

$$y = -16x + 16$$

(A1)

(7)

Alternative for $\frac{dx}{dy}$
 $\frac{1}{4}$

Use $t = \frac{1}{4}$ to calculate x

$$x = \frac{72}{y^2} - \frac{3}{y}$$

ACF CSO

(M1)

$$\frac{dx}{dy} = -\frac{144}{y^3} + \frac{3}{y^2}$$

(A1)

$$\left(\frac{dx}{dy} = \frac{3y-144}{y^3} \right)$$

Correct expression for x from candidate's implicit equation
 and attempt derivatives

(A1)

$$(ii) \quad y = -16 \times \frac{3}{2} + 16 = -8$$

Substitute $x = \frac{3}{2}$ into candidate's tangent; calculate y

M1

$$\frac{3}{2}(-8)^2 + 3 \times (-8) = 96 - 24 = 72$$

$y = -8$ used to verify 72

A1

2

[11]

Q6.

$$(a) \quad (i) \quad \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{6\cos 2\theta}{-2\sin \theta}$$

condone coefficient errors

M1

A1

$$= \frac{6(1 - 2\sin^2 \theta)}{-2\sin \theta}$$

Use $\cos 2\theta = 1 - 2\sin^2 \theta$

$$= 6\sin \theta - 3\operatorname{cosec} \theta$$

$$a = 6 \quad b = -3$$

m1

A1

4

$$(ii) \quad \theta = \frac{\pi}{6} \quad \frac{dy}{dx} = 6 \times \frac{1}{2} - 3 \times 2 = -3$$

$\theta = \frac{\pi}{6}$ substituted into their $\frac{dy}{dx}$ and evaluated

B1ft

$$\text{gradient normal} = \frac{1}{3}$$

ft $\frac{dy}{dx}$, provided non-zero

(b)

$$\left. \begin{aligned} y &= 6 \sin \theta \cos \theta \\ &= (\pm) 6 \sqrt{1 - \cos^2 \theta} \times \cos \theta \\ &= (\pm) 6 \sqrt{1 - \left(\frac{x}{2}\right)^2} \times \left(\frac{x}{2}\right) \end{aligned} \right\}$$

Correct expansion of $\sin 2\theta$ and use $x = 2\cos\theta$ to eliminate θ

M1

Correct elimination of θ

A1

$$y^2 = \frac{9}{4} x^2 (4 - x^2)$$

$$p = \frac{9}{4} \text{ OE and } (4 - x^2) \text{ shown}$$

A1

Alternative: using verification

$$y^2 = 9 \sin^2 2\theta = 36 \sin^2 \theta \cos^2 \theta$$

must be squared

(M1)

$$x^2(4 - x^2) = 4 \cos^2 \theta \times 4 \sin^2 \theta$$

(A1)

$$p = \frac{9}{4}$$

OE

$$\text{or } y^2 = \frac{9}{4} x^2 (4 - x^2)$$

(A1)

Q7.

(a) (i) $\frac{dx}{dt} = 3e^t \quad \frac{dy}{dt} = 2e^{2t} + 2e^{-2t}$

Both derivatives attempted and one correct

M1

Both correct

A1

$$t = 0 \quad \text{gradient} = \frac{4}{3}$$

$$\text{cso} \quad \text{Condone} \quad \frac{dy}{dx} = \frac{4}{3}$$

A1

3

(ii) $y^{\frac{4}{3}}(x-3)$ oe
ft on non-zero gradient

B1ft

1

(b) $e^{2t} = \frac{x^2}{9}$ or $9e^{2t} = x^2$ or $e^t = \frac{x}{3}$ or $e^{2t} = \left(\frac{x}{3}\right)^2$

or $t = \ln\left(\frac{x}{3}\right)$ or $2t = \ln\left(\frac{x^2}{9}\right)$

M1

$$y = \frac{x^2}{9} - \frac{9}{x^2}$$

Equation required

A1

2

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[6]

Q8.

(a) (i) $\left(\frac{dx}{d\theta}\right) = -6\sin 2\theta, \quad \left(\frac{dy}{d\theta}\right) = -2\sin \theta$
 $\left(\frac{dx}{d\theta}\right) = p\sin \theta$ or $r\sin \theta \cos \theta$
 $\left(\frac{dy}{d\theta}\right) = q\sin \theta$

M1

Both correct.

A1

$$\frac{dy}{dx} = \frac{-2 \sin \theta}{-6 \sin 2\theta}$$

$$\frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

Use chain rule ; condone one slip

M1

$$= \frac{2 \sin \theta}{6 \times 2 \sin \theta \cos \theta} = \frac{1}{6 \cos \theta}$$

$k = 6$ must come from correct working seen **AG**

A1

4

(ii) $\theta = \frac{\pi}{3} \quad m_T = \frac{1}{3}$

ft on k $\left(\frac{1}{k \times \frac{1}{2}} \right)$
 k need not be numerical

B1ft

$$m_N = -3$$

ft on m_T

B1ft

$$(x, y) = \left(-\frac{3}{2}, 1 \right)$$

B1

Normal $y - 1 = -3 \left(x + \frac{3}{2} \right)$

CAO; any correct form, ISW.

$$2y + 6x + 7 = 0$$

B1

4

(b) $\sin^2 x = \frac{1}{2} (1 - \cos 2x)$

$p + q\cos 2x$; Allow different letters for x or mixture eg θ even for A1 and the following A1ft

M1

A1

$$\int p \, dx = px \quad \int q \cos 2x = \frac{1}{2} q \sin 2x$$

Both integrals correct;
ft on p and q

A1ft

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x \, dx = \left[\frac{x}{2} - \frac{1}{4} \sin 2x \right]$$

$$= \left(\frac{\pi}{8} - \frac{1}{4} \right) - \left(-\frac{\pi}{8} - \left(-\frac{1}{4} \right) \right)$$

Correct use of limits;

$$F\left(\frac{\pi}{4}\right) - F\left(-\frac{\pi}{4}\right) \text{ or } 2F\left(\frac{\pi}{4}\right)$$

$$F(x) = px + r \sin 2x \text{ and } \sin \frac{\pi}{2}, \sin \left(-\frac{\pi}{2}\right)$$

must be evaluated correctly for m1

m1

$$= \frac{\pi}{4} - \frac{1}{2}$$

CSO OE ISW

A1

5

Alternative

$$\int \sin^2 x \, dx = -\sin x \cos x - \int -\cos x \cos x \, dx$$

Use parts; condone sign slips

(M1)

$$= -\sin x \cos x + \int 1 - \sin^2 x \, dx$$

Use $\cos^2 x = 1 - \sin^2 x$

(m1)

$$2 \int \sin^2 x dx = -\sin x \cos x + x$$

(A1)

$$2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = G\left(\frac{\pi}{4}\right) - G\left(-\frac{\pi}{4}\right)$$

Correct use of limits

(m1)

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = \frac{\pi}{4} - \frac{1}{2}$$

(A1)

(5)

[13]

Q9.

(a) (i) $\sin 2\theta = 2 \sin \theta \cos \theta$

B1

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

OE condone use of x etc, but variable must be consistent

B1

2

(ii) $\sin \theta = \frac{4}{5} \Rightarrow \sin 2\theta = 2 \times \frac{4}{5} \times \frac{3}{5} = \frac{24}{25}$

AG

Use of 106.26° B0

B1

or

$$2 \times \sin\left(\cos^{-1}\frac{3}{5}\right) \times \frac{3}{5}$$

$$\cos 2\theta = \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}$$

$$-0.28$$

B1

2

(b) (i) $\frac{dx}{d\theta} = 6 \cos 2\theta$, $\frac{dy}{d\theta} = -8 \sin 2\theta$

Attempt both derivatives. ie $p \cos 2\theta$

M1

Both correct. $q \sin 2\theta$

A1

$$\frac{dy}{dx} = -\frac{4 \sin 2\theta}{3 \cos 2\theta} \quad \text{ISW}$$

CSO OE

A1

3

(ii) $P\left(\frac{72}{25}, -\frac{28}{25}\right)$

$(2.88, -1.12)$

B1F

Gradient = $-\frac{4}{3} \times -\frac{24}{7}$

Their $\frac{q \sin 2\theta}{p \cos 2\theta}$ **or** $\frac{p \cos 2\theta}{q \sin 2\theta}$

must be working with rational numbers

M1

Tangent $y + \frac{28}{25} = \frac{32}{7} \left(x - \frac{72}{25}\right)$ ISW

Any correct form.

$$7y = 32x - 100$$

Fractions in simplest form

Equation required

A1

Q10.

(a) $\frac{dx}{dt} = -3 \quad \frac{dy}{dt} = 6t^2$

Both derivatives correct; PI

B1

$$\frac{dy}{dx} = -\frac{6t^2}{3}$$

Correct use of chain rule

M1

$$= -2t^2$$

CSO

A1

3

(b) $t = 1 \quad m_T = -2 \quad m_N = \frac{1}{2}$

Substitute $t = 1$ $m_N = -\frac{1}{m_T}$

M1

F on gradient; $m_T \neq \pm 1$

A1F

Attempt at equation of normal using $(x, y) = (-2, 3)$

Condone one error

M1

Normal has equation $y - 3 = \frac{1}{2}(x + 2)$

CSO; ACF

A1

4

(c) $t = \frac{1-x}{3} \quad \text{or} \quad t = \sqrt[3]{\frac{y-1}{2}}$

Correct expression for t in terms of x or y

M1

$$y = 1 + 2\left(\frac{1-x}{3}\right)^3$$

ACF

A1

2

[9]

Q11.

(a) $\frac{dx}{dt} = \frac{1}{t^2} \quad \frac{dy}{dt} = 1 - \frac{1}{2t^2}$

B1B1

$$\frac{dy}{dx} = \frac{1 - \frac{1}{2t^2}}{-\frac{1}{t^2}} \left(= \frac{2t^2 - 1}{-2} \right)$$

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$
 Their $\frac{dy}{dt}$; condone 1 slip

CSO; ISW

M1

A1

Alternative

$$y = \frac{1}{x} + \frac{x}{2}$$

(B1)

$$\frac{dy}{dx} = -\frac{1}{x^2} + \frac{1}{2}$$

(B1)

Substitute $x = \frac{1}{t}$

(M1)

$$\frac{dy}{dx} = -t^2 + \frac{1}{2}$$

CSO

(A1)

4

(b) $t = 1 \quad \frac{dy}{dx} = \frac{1}{2}$

Substitute $t = 1$ in $\frac{f(t)}{g(t)} \neq k$

M1

$$m_T = -\frac{1}{2} \Rightarrow m_x = 2$$

F on $m_T \neq 0$; if in $t \rightarrow$ numerical later

B1F

$$(x, y) = (1, \frac{3}{2})$$

$$Pl \frac{3}{2} = m(x1) + c$$

B1

$$(y - \frac{3}{2}) = 2(x - 1) \text{ or } y = 2x + c, c = -\frac{1}{2}$$

ISW, CSO (a) and (b) all correct

A1

4

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$$y = \frac{1}{1} + \frac{1}{2} \times \frac{1}{t}$$

(c)

*Attempt to use $t = \frac{1}{x}$ to eliminate t
t, or equivalent*

M1

$$= \frac{1}{x} + \frac{x}{2}$$

A1

$$2xy = 2 + x^2 \Rightarrow x^2 - 2xy + 2 = 0$$

Correct algebra to AG with $k = 2$

allow $k = 2$ stated $k = 2$,

no working or from $\left(1, \frac{1}{3}\right): 0/3$

A1

Alternative

or

$$\left(\frac{1}{t}\right)^2 - 2\left(\frac{1}{t}\right)\left(t + \frac{1}{2t}\right) \quad \left| \quad xy = \frac{1}{t}\left(t + \frac{1}{2t}\right) \right.$$

$$= -2 \quad \left| \quad = 1 + \frac{x^2}{2} \right.$$

Substitute and multiply out

(M1)

Eliminate t

(A1)

$$\Rightarrow x^2 - 2xy + 2 = 0$$

Conclusion, $k = 2$

(A1)

3

[11]

Q12.

$$t = \frac{1}{2} \quad x = 2 \times \frac{1}{2} + \frac{1}{\left(\frac{1}{2}\right)^2} \quad y = 2 \times \frac{1}{2} - \frac{1}{\left(\frac{1}{2}\right)^2}$$

(a) (i)

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$$x = 5$$

$$y = -3$$

A1

2

$$(ii) \quad \frac{dy}{dt} = 2 + 2t^{-3} \quad \frac{dx}{dt} = 2 - 2t^{-3}$$

2 and $\frac{d}{dt}\left(\frac{1}{t^2}\right)$ attempted in both derivatives

M1A1

$$t = \frac{1}{2} \quad \frac{dy}{dx} = \frac{2 + \frac{2}{1/8}}{2 - \frac{1}{8}} = -\frac{9}{7}$$

use chain rule; expressions can be in terms of t or evaluated

M1

CAO or any equivalent fraction (not decimals)

A1

$$y + 3 = -\frac{9}{7}(x - 5)$$

ft on, x y and gradient

if $y = mx + c$ used, c must be found correctly and the equation must be rewritten

B1F

5

(b) $x - y = \frac{2}{t^2} \quad x + y = 4t$

either correct expression

or both of $x - y = 4t$ and $x + y = \frac{2}{t^2}$

M1

$$\frac{2}{(x - y)} = \left(\frac{x + y}{4} \right)^2$$

eliminate t

M1

$$32 = (x - y)(x + y)^2$$

$$\text{or } (x - y)(x + y)^2 = \frac{2}{t^2} \times (4t)^2 = 32$$

$k = 32$ alone, no marks

A1

3

[10]

Q13.

(a) (i) $\frac{dx}{dt} = 2, \quad \frac{dy}{dt} = -8t$
 CAO

B1, B1

2

(ii) $\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{-8t}{2} = -4t$

M1

*Chain rule in correct form
 ft on sign coefficient errors (**not** power of t)*

A1F

2

(b) $m_T = -4, \quad m_N = \frac{1}{4}$

ft on $\frac{dy}{dx}$ if f(t)

B1F,
B1F

$x = 3 \quad y = -3$

$\frac{y+3}{x-3} = \frac{1}{4} \Rightarrow \frac{y-3}{x-3} = \frac{1}{4}$

Use candidate's (x, y) and m_N

M1

Any correct form; ISW; CAO

A1

4

(c) $t = \frac{x-1}{2}$

M1

$y = 1 - 4 \left(\frac{x-1}{2} \right)^2$

*Substitute for t
 Simplification not required but CAO
 Or equivalent methods / forms:*

$$y = 2x - x^2, \quad t^2 = \frac{1-y}{4},$$

$$\left(\frac{x-1}{2}\right)^2 = \frac{1-y}{4}$$

M1A1

3

[11]

Q14.

(a) (i) $\frac{dx}{d\theta} = -\sin\theta \quad \frac{dy}{d\theta} = 2\cos 2\theta$

B1 B1

2

(ii) $\frac{dy}{dx} = -\frac{2\cos 2\theta}{\sin\theta}, \quad \frac{dy}{dx} = -\frac{2\cos\frac{\pi}{3}}{\sin\frac{\pi}{6}} = -2$

use chain rule $\frac{dy}{d\theta}$ and $\frac{dx}{d\theta}$
 substitute $\theta = \frac{\pi}{6}$

M1

A1

2

(b) $y = 2\sin\theta\cos\theta = 2\sqrt{1-\cos^2\theta}\cos\theta$
 use $\sin 2\theta = 2\sin\theta\cos\theta$

B1

use $\sin^2\theta = 1 - \cos^2\theta$

B1

$$y = 2\sqrt{1-x^2}x$$

$\sin\theta, \cos\theta$ in terms of x

M1

$$y^2 = 4x^2(1-x^2)$$

all correct CSO

A1

4

Alt

$$y^2 = \sin^2 2\theta = (2\sin \theta \cos \theta)^2$$

use of double angle formula

(B1)

$$= (4)\sin^2 \theta \cos^2 \theta = (4)(1 - \cos^2 \theta) \cos^2 \theta$$

use of $s^2 + c^2 = 1$ to eliminate $\sin \theta$

(B1)

$$= (4)(1 - x^2)x^2$$

Substitute $\cos \theta$ for x

(M1)

$$= 4(1 - x^2)x^2$$

CSO

(A1)

(4)

[8]

Q15.

(a) $\frac{dy}{dt} = \frac{-2}{t^2} \quad \frac{dx}{dt} = -4$

M1A1

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{1}{\frac{dx}{dt}} = \frac{1}{2t^2}$$

Use chain rule

m1

Follow on use of chain rule (if $f(t)$)

A1F

Or eliminate t : M1 $y = f(x)$ attempt to differentiate M1A1 chain rule

A1F reintroduce t

4

(b) $t = 2 \quad m_T = \frac{1}{8}$

follow on gradient (possibly used later)

B1F

$$x = -5 \quad y = 2$$

B1

$$y - 2 = \frac{1}{8}(x + 5)$$

Their (x, y), m

M1

$$x - 8y + 21 = 0$$

Ft on (x, y) and m

A1F

4

(c) $x - 3 = -4t \quad y - 1 = \frac{2}{t}$
PI

M1

$$(x - 3)(y - 1) = -4t \times \frac{2}{t} = (-8)$$

Attempt to eliminate t

M1

AG convincingly obtained

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A1

3

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[11]

Q16.

(a) $t = \frac{1}{2} \quad x = 3 \quad y = 2$

B1B1

2

(b) $t = \frac{1}{y} \quad x = 2\frac{1}{y} + y$

Attempt to eliminate t
SC verification using t 1/2

M1

$$xy = 2 + y^2 \quad xy - y^2 = 2$$

AG; convincingly found

A1

2

(c) Implicit differentiation: $x \frac{dy}{dx} + y, -2y \frac{dy}{dx} = 0$

Attempt at equation with $\frac{dy}{dx}$ but not " $\frac{dy}{du} = \dots$ "

M1
A1A1

RHS = 0

B1

At (3, 2) $3 \frac{dy}{dx} + 2 - 4 \frac{dy}{dx} = 0$

Use of (3, 2)

m1

$$\frac{dy}{dx} = 2$$

AG; convincingly obtained

A1

6

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{1}{\frac{dx}{dt}} = -\frac{1}{t^2} \frac{1}{2 - \frac{1}{t^2}}$$

Parametric differentiation:

Attempt chain rule, PI

(M1)
(A1A1)

$$\frac{1}{t^2} = y^2$$

Or sub $t = \frac{1}{2}$; -4 in numerator is sufficient for this

(B1)

$y = 2$ $\frac{dy}{dx} = \frac{-4}{2-4} = 2$

AG; convincingly obtained

(M1A1)

(6)

OR

$$x = y + 2y^{-1} \quad \text{M1}$$

$$\frac{dx}{dy} = 1 - 2y^{-2} \quad \text{M1A1A1}$$

$$= 1 - \frac{1}{2} = \frac{1}{2} \quad \text{B1} \rightarrow \frac{dy}{dx} = 2 \quad \text{A1}$$

[10]



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